# CS-002 Layup Schedule Specification & Recording

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| **Doc ID** | CS-002 |
| **Title** | Layup Schedule Specification & Recording |
| **Revision** | D |
| **Status** | Draft, pending Lead signature |
| **Effective date** | 2026-08-28 |
| **Supersedes** | Master Tracker “Layup Schedule” shorthand as the only stack record |

### Approvals

| Role | Name | Date |
|---|---|---|
| Author | Simon Starbuck (drafted with Claude, SN6 Resources) | 2026-07-12 |
| Reviewer | simon (reviewer agent) | |
| Approver (Composites Lead) | | |

### Revision history

| Rev | Date | Author | Description of change |
|---|---|---|---|
| A | 2026-07-12 | Claude | Initial issue |
| B | 2026-07-28 | Claude | Language pass: fewer em dashes for rhythm variety, no content change. |
| C | 2026-08-03 | Claude | §7 gains the coupon suffix convention and the rule that coupon IDs are written on the panel before the first cut (CS-001 §7.6, CS-013 §4.1). |
| D | 2026-08-28 | Claude | Engineering pass (CS-000 Rev C conventions). Figure 1 draws the §7.3 example stack in section, so “P1 is the mold side” stops being a convention people have to hold in their heads at the cutting table. No rule changed. |

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## 1. Purpose

In SN5 our stacks lived in the tracker as shorthand (“6X 195 + CORE”, “195 88 .25 NOMEX”). That’s the *intent*, not what actually got laid. Orientations, core placement, where the copper mesh went, what we substituted when we ran out of cloth: all of it lived in people's heads. That's how the dashboard got remade 3+ times against a spec that existed nowhere (PP-04), how the UT stack was still up in the air three weeks before layup (PP-07), and why nobody could answer "what stack did the seat use last year?" (PP-09). This doc is how we write a stack down before we build, and record what we actually did after.

## 2. Scope

Every laminated part: infusion, wet layup, foam-wrapped, and forged CF (mass/ratio spec instead of ply table). Applies to R&D coupons too, since untraceable test data is noise.

## 3. References

| Ref | Document | Where |
|---|---|---|
| R1 | PP-04, PP-07, PP-09 RCAs | FEB-COMP-PP-001 |
| R2 | CS-013 Work Orders (where stacks are recorded) | CS-013 |
| R3 | Rohacell IG-F TDS (core naming example) | `04 Datasheets/ROHACELL_IG_F_2022_April_EN_243` |
| R4 | Materials Testing & Recommendation (SN5 coupon data) | `SN5 Composites/Documentation/Materials Testing & Recommendation.docx` |

## 4. Definitions

- **Stack / layup schedule**: the full ordered list of plies from mold surface (P1) outward, including cores, meshes, and inserts.
- **Orientation**: warp direction relative to the part's marked 0° datum. For woven cloth use 0/90 or ±45; for spread-tow/UD state the fiber direction.
- **Cloth naming**: `<gsm> <weave>` from the team's stock: 88 spread-tow, 195 twill, 220 twill, 670 twill (Sigmatex); always gsm, never "the thick one".
- **Stack freeze**: the signed decision that the stack will not change without a WO revision.

## 5. Equipment & Materials

None; this is a specification standard. The stack table lives in the part's work order (R2).

## 6. Safety

No physical hazards; specification standard. Shop-floor safety lives in the process standards it feeds (CS-006, CS-007).

## 7. Procedure

### 7.1 Specify (before any manufacturing)

1. The part's Manufacturing Engineer drafts the stack as a ply table (format in 7.3), from mold surface outward.
2. Justify the choice in one or two lines per decision: coupon data (R4), the SN5 as-built for a similar part (its WO), or an explicit assumption flagged for testing. New unproven stacks on schedule-critical parts require a coupon or test panel first.
3. Include function plies explicitly: sacrificial 88 spread-tow outer layers for sanding protection, copper mesh position and overlap for grounded parts (CS-010), core chamfers/grooves.

4. State the target: cured thickness (mm), mass (g), and any acceptance measurement (e.g. grounding Ω per CS-010).

### 7.2 Freeze

1. **The stack freezes before mold machining starts.** Mold geometry (offsets, flange height) can depend on stack thickness, and machine slots are too scarce to cut twice (PP-03, PP-06). For moldless flat panels, freeze before cloth is cut.
2. Freeze = ME/RE and the Composites Lead (plus the requesting subteam's lead for cross-team parts, the dashboard lesson, PP-04) initial the WO's stack table.
3. Every stack-freeze date appears as a deadline in the season timeline, reviewed at the Monday meeting (PP-07). A frozen-stack change requires a WO revision: quick to do, but visible and counted.

### 7.3 Ply table format

| Ply | Material | Orientation | Coverage | Notes |
|---|---|---|---|---|
| P1 (mold side) | 88 spread-tow | 0/90 | full | sacrificial/finish |
| P2 | 195 twill | ±45 | full | |
| P3 | Cu mesh | n/a | zones A–B | grounding, 25 mm overlap onto P2 |
| P4 | Rohacell 31 IG-F 3 mm | n/a | core zone | chamfer 20°, grooves per CS-006 |
| P5–P6 | 195 twill | 0/90, ±45 | full | |
| P7 (bag side) | 88 spread-tow | 0/90 | full | |

(Example only: the table above is the *format*, not a recommended stack.)

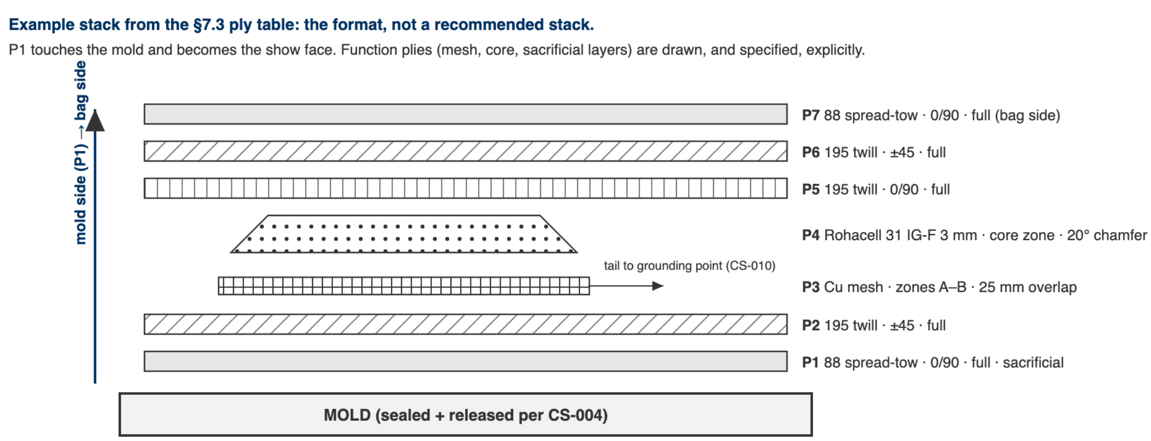


Figure 1: The §7.3 example stack drawn in section. P1 touches the mold and becomes the show face; function plies are drawn, and specified, explicitly.

### 7.4 Record as-built (during/after layup)

1. During layup, the RE ticks each ply as laid and notes any deviation in the WO (substituted cloth, extra ply, mesh moved). No deviation is too small to note: "we were out of 195 so used 220" is exactly the data that explains a mass or thickness surprise later.

2. Photograph the stack at: cores/mesh placed, and bagged-before-infusion. Photos attach to the WO steps.
3. After cure, record actual mass and thickness next to targets in the WO qualityChecks.

**Coupons**

A coupon is `PNL-SNx-###-Cnn`: its panel's ID, plus the cut sequence left to right. It is a suffix rather than its own record because a coupon is never re-parented (it was physically cut from exactly one panel, and that is true forever) and because there are dozens per batch, all of them in two pieces by the end of the week.

1. **Write every coupon ID on the panel before the first cut**, across the cut lines, so each piece inherits a legible fragment. After the saw there is no way to recover which piece was which.
2. Name the raw data file after the coupon: `PNL-SN6-006-C03.csv`. That one filename then carries the panel, the stack, the session and the material lots, through one link each, none of them retyped.

   This is written against a real failure. The SN5 tensile set is seventeen CSVs named `TensileTesting_12_8_2025_<n>.csv` whose contents are Time, Displacement and Force with no header block and no specimen identity of any kind. The mapping from specimen to layup exists only in a Word table that contradicts itself. Untraceable test data is noise (§2).

## 8. Quality checks / acceptance criteria

| Check | Target | Method | Record where |
|---|---|---|---|
| Stack frozen before mold machining / cloth cut | Initials + date present | WO blocker step | WO |
| As-built matches frozen stack | Any deviation noted, none silent | RE self-check + photo review | WO |
| Mass vs target | Within stated tolerance (default ±10%) | Scale | WO qualityChecks |
| Thickness vs target | Within stated tolerance | Calipers at 3+ points | WO qualityChecks |

## 9. Records

The WO ply table (spec + as-built + deviations + photos) is the permanent record. The Master-Tracker-style shorthand may still be used in dashboards/summaries, but it is a *view* of the WO, never the source of truth.